Lime Harvesting Robot Challenge

This competition aims to encourage students to apply creativity and engineering skills in designing a functional robot. The task focuses on developing an automated system that can identify and pick ripe limes while leaving unripe ones on the tree. To ensure fairness, clear competition rules and technical specifications have been established for all teams. Evaluation will be conducted based on a structured marking rubric that combines robot functionality and task performance.

Figure 1: A plastic lime plant in a pot with artificial ripe (yellow) fruits, used as the target model for the robot-picking task.

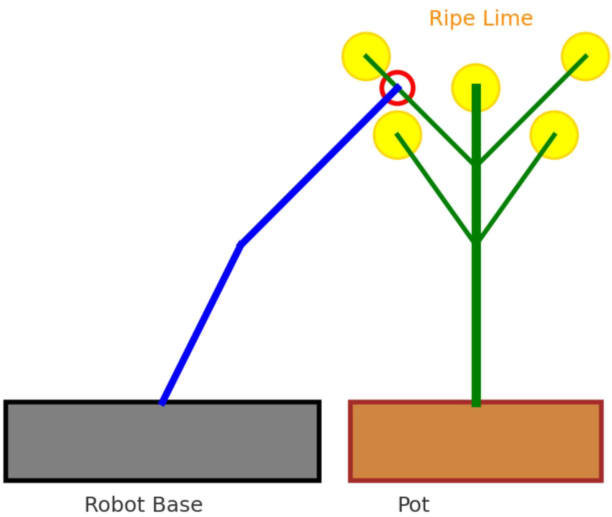

Figure 2: Conceptual illustration of the lime-picking robot. The robot base is positioned minimum 50 cm from the pot, with an arm and gripper mechanism reaching towards a ripe lime. *This is only a conceptual representation; the final design will depend on the creativity of each team.*

Competition Rules

1. **Robot Design**
   - Each team must design a robot capable of picking ripe limes (*see Figure 2*).
   - Ripe limes (yellow in color) must be picked and placed at the designated area.
   - Unripe limes (green in color) must remain on the tree.
2. **Plant and Robot Setup**
   - The base of the plant pot is positioned minimum 50 cm above the floor (*see Figure 1*).
   - The distance between the robot base and the pot is 50 cm.
   - The robot's working space must be at least 30 cm (width) × 30 cm (height) × 30 cm (length).
3. **Team Task**
   - Each team must pick **3 ripe limes**.
   - If unable to pick 3, then 2 or 1 ripe lime is acceptable.
4. **Winner Determination**
   - Evaluation will be based on two criteria:
     a) **Functionality of the Robot** – how well the robot performs the task (accuracy in picking ripe limes, ability to avoid unripe limes, and overall reliability of movement).
     b) **Task Performance** – number of ripe limes successfully picked. If the number is the same, the team that completes the task in the shortest time will be declared the winner.
5. **Overall Winner**
   - The overall winner will be determined by combining both criteria: **robot functionality** and **task performance**.

Competition Rubrics

Criteria	Excellent (8-10 points)	Good (6-7 points)	Fair (4-5 points)	Poor (0-3 point)
1. Fulfilling design specification				
Picking Success Rate (5)	Successfully picks 3 ripe limes.	Successfully picks 2 ripe limes.	Picks only 1 ripe lime.	Fails to pick any ripe lime.
Placement Success Rate (5)	Successfully places 3 ripe limes into designated area.	Successfully places 2 ripe limes into designated area.	Places only 1 ripe lime into designated area.	Fails to place any ripe lime.
2. Innovativeness				
Accuracy in Visual Servoing (10)	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 3 ripe limes.	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 2 out of 3 ripe limes.	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 1 out of 3 ripe limes.	Fails to demonstrate precise alignment between visual detection and end-effector.
3. Quality of work				
Design & Innovation (10**)**	Product shows excellent quality of work in: i. Human–Machine Interface (HMI) is **properly designed** with **user-friendliness** as a top priority. ii. The prototype is housed within a **well-designed enclosure or casing** that offers adequate **protection** and presents a **professional appearance**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, professionally labelled and documented** in diagrams. iv. **Maximize** the utilization of **printed circuit board (PCB) or strip / donut board**. **Proper connectors** and **wires** are studied, selected and used.	Product shows good quality of work in: i. Human–Machine Interface (HMI) is **user-friendly**, but there may be **minor** areas where further **improvement** is possible. ii. The prototype is housed within a **well-designed enclosure or casing**, which may need a **minor improvement**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, labelled and documented**, which may **require minor refinements** for greater clarity. iv. **Maximize** the utilization of **printed circuit board (PCB) or strip / donut board**. But the **connectors** and **wires** are used **without proper study**.	Product shows fair quality of work in: i. Human–Machine Interface (HMI) exhibits **some user-friendliness**. ii. The prototype is housed within an **enclosure or casing**, that **lacking** of either adequate **protection** functionality or profesional **appearance**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, labelled and documented**, which may **require major refinements** for greater clarity. iv. The use of breadboard together with **printed circuit board (PCB) or strip / donut board**. **Jumper wires** are **mostly used**.	Product shows poor quality of work in: i. Human–Machine Interface (HMI) **lacks user-friendliness**. ii. The prototype is **not protected by an enclosure or casing**, that may affect the safety of the end-user and reliability of the prototype. iii. Wiring, piping and tubing (if applicable) are **disorganized** and **without a proper label or documentation**. iv. The use of **breadboard** and **jumper wires**.
4. Live demo				
Completion Time for Two Picking Attempts (10)	Completes the task within the given time.	Completes the task longer than the given time.	Manage to successfully pick and place at least 1 ripe.	Unable to complete the task.

<u>Terms and conditions</u>

1. Each team will be given **5 minutes** to complete the task, with a **maximum of two trials** allowed. The **total number of harvested lemons** from both trials will be **accumulated** for scoring.
2. The setup will consist of **three ripe** and **three unripe** lemons, which will be **placed randomly** during the competition.
3. The **stem** of each lemon will be made using **rolled brown paper** measuring approximately **1.5 cm × 3 cm**. Participants may select the **type of paper** based on **suitability** for their gripping or detection mechanism. Refer to Figures 3 and 4.
4. **Marks will be deducted by 50%** if a lemon is harvested **without proper cutting** of the stem.
5. The **judge's decision is final**.

Figure 3: The left-hand side shows the correctly rolled paper stem (the color should be brown). The right-hand side is incorrect, as the paper has not been rolled properly.

Figure 4: Multiple viewing angles of the lemons attached to the tree.

Frequently Asked Questions (FAQs)

Q1: How are errors like picking or dropping unripe fruit penalized?

A1: The "Picking Success Rate" is calculated using the net successful picks. This is determined by subtracting any incorrect harvests (picking or causing an unripe lemon to fall) from the total number of successfully picked ripe lemons.

Example:
Ripe Lemons Picked: 2
Unripe Lemons Dropped/Picked: 1
Net Successful Picks: 2 - 1 = 1 (This value is used for scoring)

Q2: How is picking ripe fruit without pruning penalized?

A2: As highlighted by the end user (Mr. Helmi), harvesting ripe fruit without proper pruning poses safety risks and may damage the fruit, tree, or even the worker. Therefore, this is treated as an incorrect harvest.

This is determined by subtracting any incorrect harvests (picking ripe fruit without pruning) from the total number of successfully picked ripe lemons.

Example:
Ripe Lemons Picked: 2
Ripe Lemons Picked Without Pruning: 1
Net Successful Picks: 2 - 1 = 1 (This value is used for scoring)

Q3: Why is Robot Quarantine Required?

A3: Robot quarantine is required to ensure fairness and integrity during the competition.

Once the robot is placed in the quarantine area, no further adjustments, programming changes, or repairs can be made. This rule ensures that all teams compete using the same conditions, without last-minute modifications that may give an unfair advantage.

Quarantine Duration:
Each robot will be quarantined for a specified duration, which will be announced before the competition begins.

Quarantine Area:
All teams must place their robot in the designated quarantine area—at least 50 cm away from the plant pot—before the competition starts.

No Access During Quarantine:
Once the robot has been placed in the quarantine area, participants are not allowed to retrieve, adjust, or touch the robot until it is their turn to compete.

Violation Consequences:
Any violation of the quarantine rules will result in the team being disqualified from that competition round and awarded 0 marks for all related rubric items.

Q4: How should a team start and reset the system during the 5-minute task window (with up to two trials allowed)?
A5: As highlighted by the end user (Mr. Helmi), the system must be user-friendly and require minimal human intervention.

Therefore, each robot must include a clearly labelled and easily accessible physical button that performs the Start / Stop / Reset functions:
- 1st Press: Starts the system for the first trial.
- 2nd Press: Stops the system (only allowed if the team decides to abort the first trial).
- 3rd Press: Resets the system, preparing it for the second trial.

During the entire 5-minute competition period, teams must not make any physical adjustments or modifications to:
- the robot hardware
- the lemon, tree, or pot
- the vision system
- any functional components

Only the Start/Stop/Reset button operations are allowed. Any violation will result in the harvesting process being terminated immediately, and the latest valid result prior to the violation will be taken as final.

Q5: What is the required placement distance for the robot?
A5: The robot must be positioned so that the distance between the center of the robot arm and the center of the plant/pot is at least 50 cm, as illustrated in Figure Q5.a.

This requirement is adapted from the actual full solution shown in Figure Q5.b, which includes both a robotic arm and a mobile base. However, your scope is limited to a fixed robotic arm, facing the lemon tree from one predefined viewing direction, as indicated within the rectangular boundary.

In Figure Q5.c, one possible layout is illustrated, showing the placement of the robot controller, power unit, and basket. The designed robot must meet the required precision and speed criteria specified in the assessment rubrics. You may select any robot controller that is suitable and sufficiently powerful for the required tasks. Since the setup involves testing a fixed robotic arm rather than a mobile platform, the system may be powered using an AC supply. The basket may be positioned at the front, side, or rear of the robot, depending on your design preferences and operational needs. For the fruit samples, you are required to prepare three ripe lemons and three unripe lemons, where the unripe lemons may be painted in any suitable shade of green.

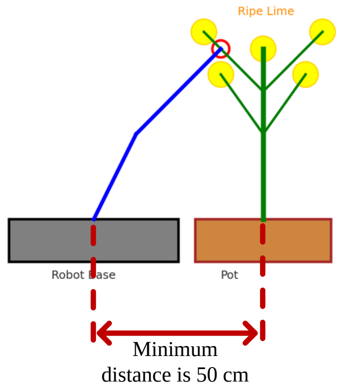

Figure Q5.a Simplified diagram for robot placement.

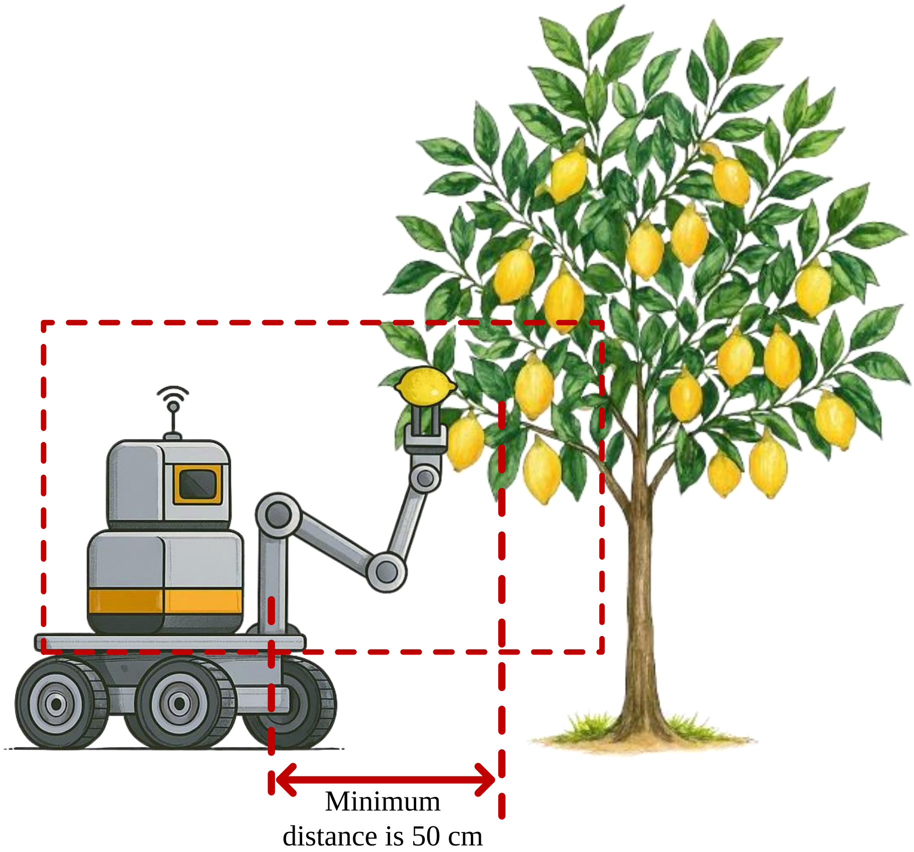

Figure Q5.b Shows the actual solution, which includes both a robotic arm and a mobile base. However, your task is limited to a fixed robotic arm facing the lemon tree from a single predefined viewpoint, as indicated within the rectangular boundary.

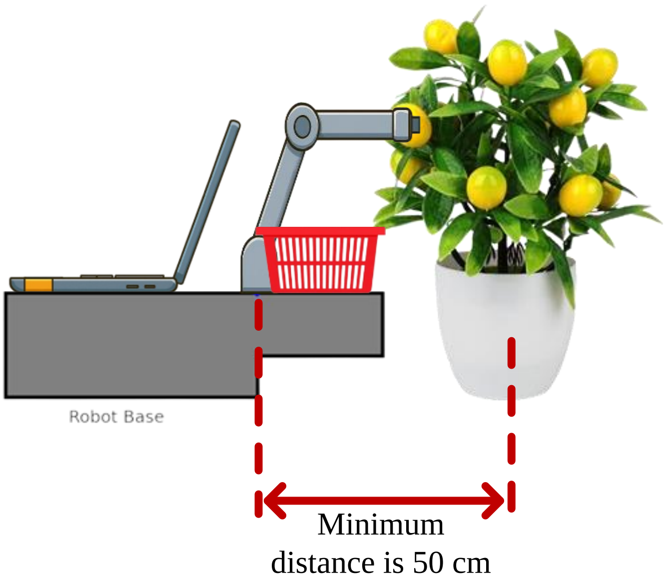

Figure Q5.c Illustrates an example arrangement of the robot, the basket, and the lemon tree.

Q6: Any modification allowed on the pot, tree and lemon?
A6: No structural modifications are permitted to the pot or the tree. Participants may, however, add external stabilizing elements—such as weights or supports placed inside the pot—provided these additions do not alter the original shape, size, or structure of the pot or tree. Please refer to Figure Q6.a.

As briefed earlier, the stem of each lemon will be fabricated using rolled brown paper with approximate dimensions of 1.5 cm × 3 cm. Participants may choose the type of paper based on its suitability for their gripping or detection mechanisms. Figure Q6.b illustrates how the 1.5 cm × 3 cm brown paper is transformed into an artificial stem.

Regarding the lemons, as previously briefed, each tree will contain three green (unripe) lemons and three yellow (ripe) lemons. Each lemon will be placed at the end of a branch using the artificial stem, at the same location where the original lemon was installed by the manufacturer. The exact arrangement of the six lemons will be determined by the judges during the competition. However, all six lemons will be positioned to face the robots, as shown in Figure Q6.c.

Please note that any violation of these rules—such as unauthorized modification to the pot or tree, or incorrect placement of the lemons—will result in corrective action by the judges. This may include swapping the lemon tree with another group, rotating or repositioning the pot, or changing the placement of the lemons. The judges' decision is final.

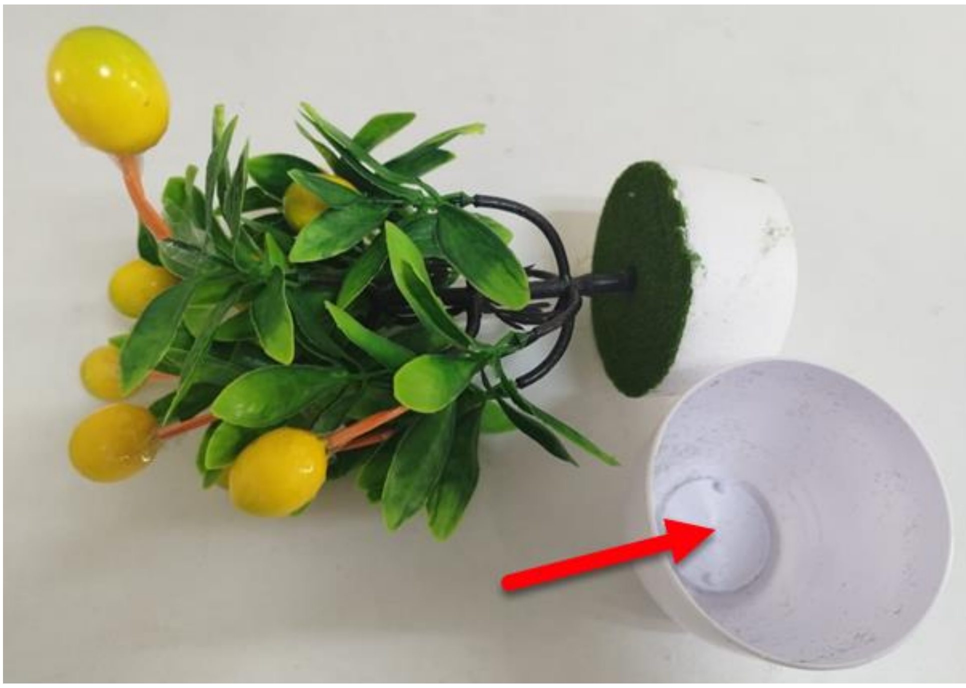

Figure Q6.a Weights or supports may be placed inside the pot (red arrow). Structural modification of the pot or tree is not allowed.

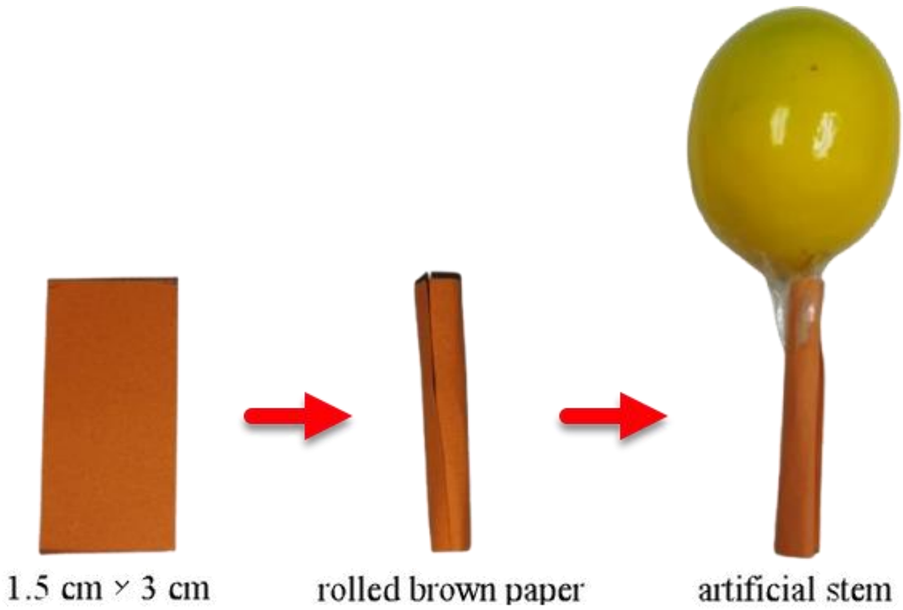

Figure Q6.b Illustration showing how a 1.5 cm × 3 cm paper strip is rolled to form an artificial lemon stem.

Figure Q6.c All six lemons are arranged to face the robot, with minimal obstruction to facilitate reliable vision detection.